



CENTRAL UNIVERSITY OF KASHMIR

Department of Information Technology

Lab Report For

“Computer Networks Lab”

“BTCS 606 L”

**SUBMITTED IN FULFILLMENT OF THE REQUIREMENTS FOR
COURSEWORK ASSIGNMENT**

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Under the Guidance of:

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Academic Year: 2025 – 2026

CERTIFICATE

This is to certify that the assignment entitled
"Lab Report For Computer networks lab"

has been successfully completed and submitted by the following students as part of their course assignment for the academic year **2025–2026**:

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This project was undertaken as an individual assignment under the guidance of Dr Shahid Sultan, and is a requirement for the internal evaluation of the subject **CN-LAB (BTCS-606L)** in the **IT Department, Central University Of Kashmir**.

Date: _____

Place: _____

Instructor

Name: _____

Designation: _____

Introduction

This technical report details the design, configuration, and validation of a multi-subnet corporate intranet environment simulated within Cisco Packet Tracer.

In modern enterprise networking, separating operational departments from core application services is essential for optimizing traffic flow, enhancing security boundaries, and managing broadcast domains.

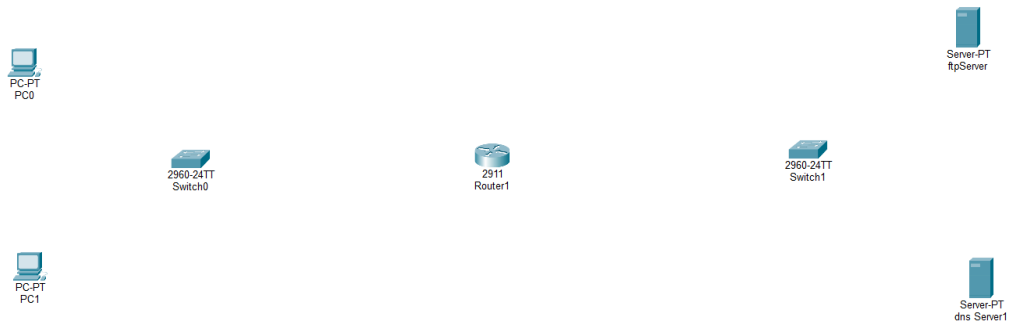
The primary objective of this laboratory project was to establish seamless Inter-VLAN routing across two distinct networks—the User Endpoints zone (192.168.1.0/24) and the Central Services cluster (10.0.0.0/8)—utilizing a Cisco 2911 Layer 3 router as the default gateway.

Beyond foundational connectivity, the project focuses on provisioning and validating application-layer architectures. This was achieved by deploying a centralized Domain Name System

(DNS) server to map human-readable domain names (ftp.nanna.com) to their respective IP addresses, alongside a secure File Transfer Protocol (FTP) server to facilitate cross-network data storage and retrieval.

Complete network integrity was evaluated and verified through end-to-end ICMP ping tests and successful remote file transmission from the client workstations.

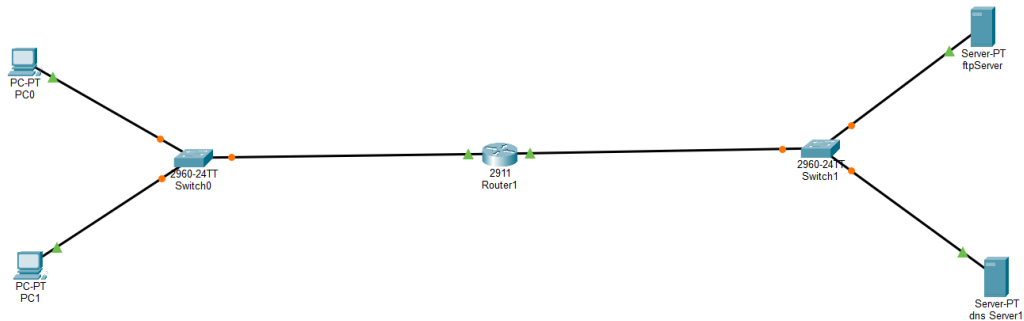
Phase 1: Hardware Placement



Open a blank workspace in Packet Tracer and drag the following devices onto the screen:

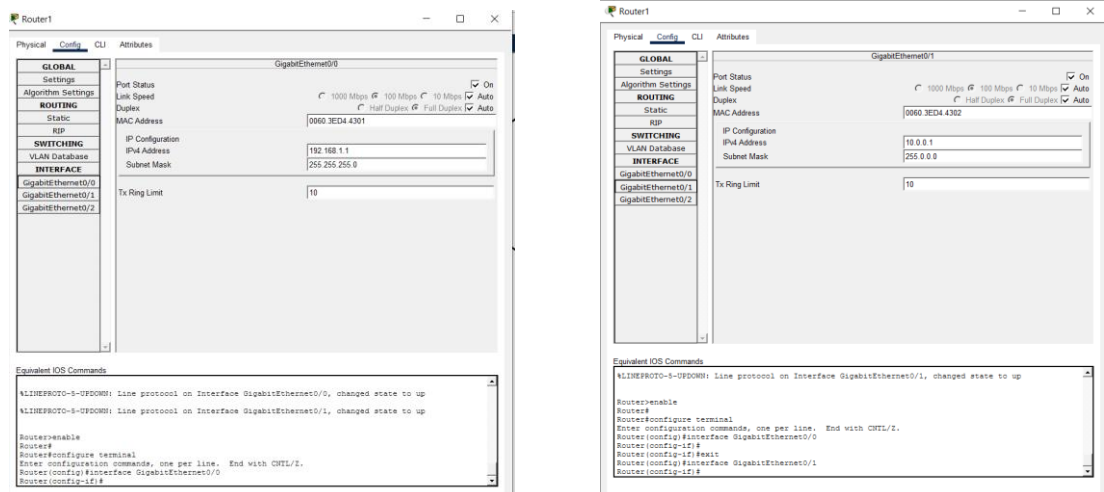
1. ***One Router:*** Select Network Devices → Routers → Choose the ***2911*** model. Place it in the center.
2. ***Two Switches:*** Select Network Devices → Switches → Choose the ***2960*** model. Place one on the left (Staff) and one on the right (DMZ).
3. ***Two PCs:*** Select End Devices → PC. Place both on the left side under Switch 1. Name them ***PC1*** and ***PC2***.
4. ***Two Servers:*** Select End Devices → Server. Place both on the right side under Switch 2. Name one
5. ***FTP Server*** and the other ***DNS Server***.

Phase 2: Cabling the Network



- 1 . Connect ***PC1*** (FastEthernet0) to ***Switch 1*** (FastEthernet0/1).
- 2 . Connect ***PC2*** (FastEthernet0) to ***Switch 1*** (FastEthernet0/2).
- 3 . Connect ***FTP Server*** (FastEthernet0) to ***Switch 2*** (FastEthernet0/1).
- 4 . Connect ***DNS Server*** (FastEthernet0) to ***Switch 2*** (FastEthernet0/2).
- 5 . Connect ***Switch 1*** (GigabitEthernet0/1) to the ***Router*** (GigabitEthernet0/0).
- 6 . Connect ***Switch 2*** (GigabitEthernet0/1) to the ***Router*** (GigabitEthernet0/1).

Phase 3: Router Configuration (The Gateway)



The router needs IP addresses on its ports to act as the gateway for your two separate networks.

1. Click on the ***Router*** and navigate to the ***CLI*** tab.
2. If it says Would you like to enter the initial configuration dialog? [yes] type **no** and hit Enter.
3. Press Enter again to get the **Router>** prompt.
4. Type the following commands exactly as written, pressing Enter after every line:

```
Router> enable
Router# configure terminal
Router(config)# interface GigabitEthernet0/0
Router(config-if)# ip address 192.168.1.1 255.255.255.0
Router(config-if)# no shutdown
Router(config-if)# exit
Router(config)# interface GigabitEthernet0/1
Router(config-if)# ip address 10.0.0.1 255.0.0.0
Router(config-if)# no shutdown
Router(config-if)# exit
```

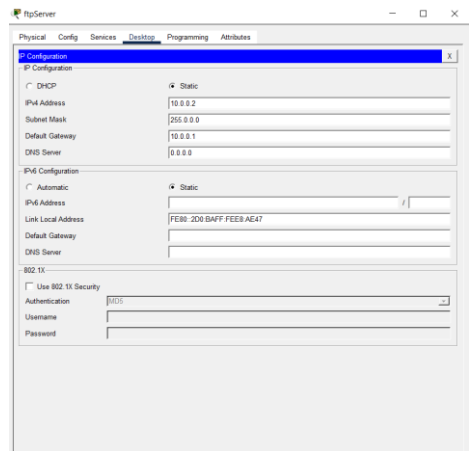
Close the router window. You should see the red connection dots turn green.

Phase4:Server Zone Configuration (Right Side)

Now, assign static IPs to the servers and turn on their respective services.

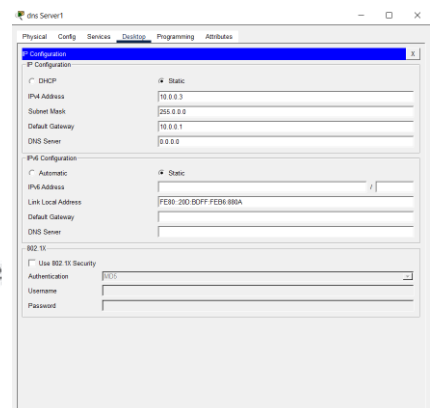
1. FTP Server Setup:

- Click ***FTP Server*** → ***Desktop*** tab → ***IP Configuration***.
- ***IPv4 Address:*** 10.0.0.2
- ***Subnet Mask:*** 255.0.0.0
- ***Default Gateway:*** 10.0.0.1 (This points to the router)
- Go to the ***Services*** tab → ***FTP***. Make sure it is ****On****.
- Under User Setup: Username **student**, Password **cisco123**.
- Check all boxes: Read, Write, Delete, Rename, List. Click ***Add***. Close window.

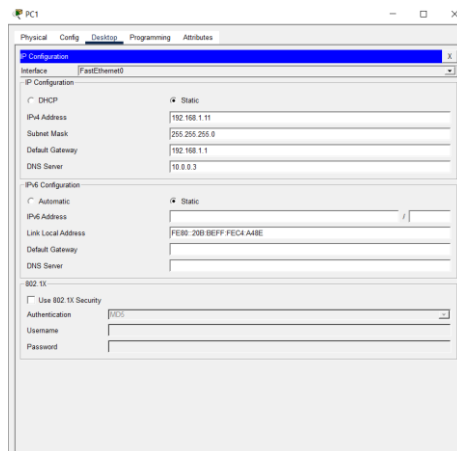


2. DNS Server Setup:*

- Click ***DNS Server*** → ***Desktop*** tab → ***IP Configuration***.
- ***IPv4 Address:*** 10.0.0.3
- ***Subnet Mask:*** 255.0.0.0
- ***Default Gateway:*** 10.0.0.1
- Go to the ***Services*** tab → ***DNS***. Make sure it is ****On****.
- Under Resource Records: Name **ftp.myproject.com**, Type **A Record**, Address **10.0.0.2**
- Click ***Add***. Close window.



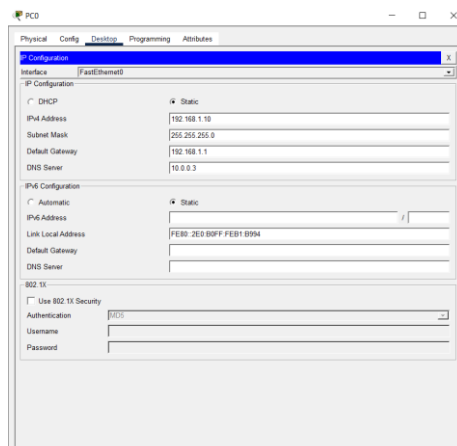
Phase 5: Staff Zone Configuration (Left Side)



Configure the client machines so they know how to reach the router and the DNS server.

1. PC1 Setup:

- Click *PC1* → *Desktop* tab → *IP Configuration*.
- *IPv4 Address:* 192.168.1.10
- *Subnet Mask:* 255.255.255.0
- *Default Gateway:* 192.168.1.1
- *DNS Server:* 10.0.0.3 (Crucial step: This points across the router to the DNS server)
- Close window.



2. PC2 Setup:

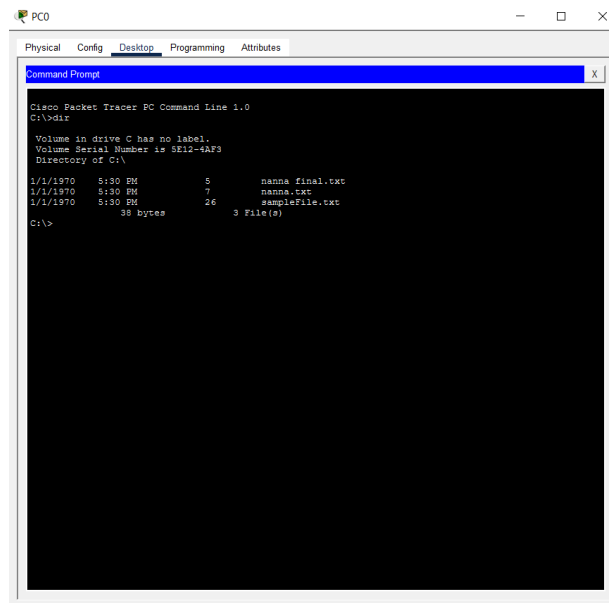
- Click *PC2* → *Desktop* tab → *IP Configuration*.
- *IPv4 Address:* 192.168.1.11
- *Subnet Mask:* 255.255.255.0
- *Default Gateway:* 192.168.1.1
- *DNS Server:* 10.0.0.3
- Close window.

Phase 6: Execution and Testing (The Experiment)

This is where you prove the architecture works.

Step A: Create a File

1. Click ***PC1*** → ***Desktop*** tab → ***Text Editor***.
2. Type some text: *"Network architecture test successful."*
3. Press **Ctrl+S**, save the file as **testfile.txt**, and close the editor.



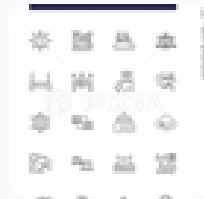
Step B: DNS & Upload Test (From PC1)

1. On ***PC1***, open the ***Command Prompt***.
2. Verify the router is passing traffic by pinging the DNS server: Type **ping 10.0.0.3**. Wait replies.
3. Connect to FTP using the domain name: Type **ftp ftp.myproject.com** and hit Enter.
4. Log in with Username **student** and Password **cisco123**.
5. Upload the file: Type **put testfile.txt**. You will see a "Transfer complete" message. Type

Step C: Download Test (From PC2)

1. Open ***PC2*** → ***Desktop*** tab → ***Command Prompt***.
2. Connect to the server: Type **ftp ftp.myproject.com** and hit Enter.
3. Log in with Username **student** and Password **cisco123**.
4. Download the file: Type **get testfile.txt**. You will see a "Transfer complete" message **quit**.
5. Type **dir** in the PC2 command prompt to view local files. You will see **testfile.txt** is PC2.

```
C:\>cmd
C:\cmd>ftp ftp.tecadadmin.net
Connected to ftp.tecadadmin.net.
220 vsFTPd 3.0.2+ <ext.1> ready...
User <ftp.tecadadmin.net:(none)>?: rahul
331 Please specify the password.
Password:
230 Login successful.
ftp>
ftp> cd uploads
250 Directory successfully changed.
ftp>
ftp>
ftp> get file1.txt
200 PORT command successful. Consider using PASV.
226 Opening BINARY mode data connection for file1.txt (2214 bytes).
```



Project Successfully Validated

This lab demonstrates the seamless integration of DNS mapping, FTP service permissions, and multi-network routing in a simulated environment.

[Any Questions?](#)

Thank You !
